

$$P(C_k \mid x) \propto P(x_1, x_2, \dots, x_n, C_k)$$

Bayes, evidence dropped

$$= P(x_1 \mid \textcolor{red}{x_2}, \dots, \textcolor{red}{x_n}, C_k) P(x_2 \mid \textcolor{red}{x_3}, \dots, \textcolor{red}{x_n}, C_k) \cdots P(x_n \mid C_k) P(C_k)$$

chain rule (exact)

$$\approx P(x_1 \mid C_k) P(x_2 \mid C_k) \cdots P(x_n \mid C_k) P(C_k)$$

naive assumption: drop the red parts

$$= P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

the Naive Bayes formula

$$\hat{y} = \arg \max_k P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

maximum a posteriori (MAP) rule